

Abdelrahman Ali

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Profile Summary

A developer with over three years of experience building financial applications, including ETL pipelines, dashboards, and market data integrations. He is now moving towards AI platform and backend engineering, building on an M.Sc. in Economics and Data Analysis (Machine Learning incl. LLMs, NLP, Big Data Management with PostgreSQL) and a thesis applying ML to EU inflation forecasting. With a B.Eng. in Electrical Engineering as well, he combines full-stack development experience with a proven ability to pick up new tools quickly.

Residence status: European Blue Card (carte bleue européenne)

Experience

European Stability Mechanism (ESM), Financial Applications Developer - Develop and deploy applications that address technical, business, and security requirements. - Develop unit tests and facilitate User Acceptance Testing (UAT). - Refactor legacy code and integrate it with newly developed applications. - Build ETL processes for the scheduled update of market data. - Write technical documentation and publish it in SharePoint and Microsoft Azure. - Use financial data APIs (Bloomberg, Refinitiv) to retrieve and process market data. - Technologies: Python Dash, Pandas, SQLAlchemy, MS SQL Server, Jenkins, Git, Selenium, unittest, SonarQube, Pylint, Bloomberg API, Refinitiv API, SharePoint.	Luxembourg May 2024 – present 2 years 5 months
European Stability Mechanism (ESM), IT Officer - Six-month traineeship during the final semester of my master’s degree, later extended to one year. - Led a transformation project migrating market data and dashboards from Tableau to Power BI. - Translated client needs into technical solutions across teams. - Developed Power Platform solutions for business process automation. - Technologies: Power BI, Power Platform, Power Automate, Git, SharePoint, change management, quantitative data analysis.	Luxembourg May 2023 – May 2024 1 year 1 month
DAL Group, Project Manager & Business Analyst - Offered a two-year contract after completing the Group’s Graduate Development Program. - Delivered business intelligence and process optimisation projects. - Managed agricultural projects end-to-end, from land preparation to harvesting and distribution. - Delivered a security system project under a tight budget and deadline. - Technologies: Project management, business analysis, Excel, MS Project, dashboard development, data analysis.	Sudan Feb 2020 – Oct 2021 1 year 9 months
DAL Group, Graduate Development Program - Two-year rotational program across Supply Chain, HR, Sales & Marketing, and Business Development. - Optimised business processes and analysed market data to build a comprehensive customer database. - Developed a three-year business model that led to the establishment of a new department.	Sudan Feb 2018 – Jan 2020 2 years

Education

- M.Sc. University of Trier & University of Bergamo (Double Degree)**, Economics and Data Analysis Germany & Italy
Oct 2021 – July 2023
- Core modules: Machine Learning (incl. LLMs), Python, Statistics, Data Analysis, Microeconomics, R, Natural Language Processing, Big Data Management (incl. PostgreSQL), Macroeconomics, International Energy Markets, Political Economics, EU Law.
 - Thesis: Predicting Inflation in the European Union Using Machine Learning.
- B.Eng. Sudan University of Science and Technology**, Electrical Engineering (Control) Sudan
- (Hons)** Sept 2012 – Sept 2017
- Core modules: Analog & Digital Electronics, Computer Applications, Electrical Circuit Theory, Microprocessor Systems, Digital Signal Processing, Digital & Fuzzy Control Systems, Digital Communications, Probability & Applied Statistics, Engineering Economics, PLCs.
 - Graduation project: Building a CNC machine for PCB manufacturing.

Skills

Languages: Python, C, C++, SQL; basic HTML, JavaScript and CSS.

Tools & Practices: Azure DevOps, Git, Jenkins, SonarQube, Pylint, unit testing, Selenium.

Data & APIs: Pandas, SQLAlchemy, REST API integration, ETL pipelines, MS SQL Server, PostgreSQL (coursework), Bloomberg API, Refinitiv API.

BI & Automation: Power BI, Power Automate, Tableau, custom automation tools.

AI/ML foundation: Machine Learning and LLMs, NLP (M.Sc. coursework); applied ML thesis on EU inflation forecasting.

Currently building (AI platform direction): FastAPI, Docker, retrieval-augmented generation (RAG), LLM-based agents — self-directed, extending the coursework foundation above.

Domain: Database management, data analysis & visualisation, change management, application design & development, code review & quality.

Languages

Arabic: Mother tongue.

English: C1 (confirmed by IELTS Academic).

French: A2 (ongoing at INL).

German: A1.